Digital Complexity: Beyond Human Understanding, Käte Hamburger Kolleg: Cultures of Research

Digital Complexity – Digital Transformation of Science and Society, Did The Computer Drive Science? Hardware Development And Digital Complexity In The 20th Century, WiSe 25/26

~~Chiara Schumann,~~ Dr. Phillip Roth, PD Dr. Ulf Hashagen

Shubham Das, 430067

03.02.2026

Protocol with Reflection (2 CP)

Formatted: Font: 12 pt, English (United States)

Dear Mr. Das,

While the content of your Protocol with Reflection is detailed and adequate, several substantial issues need to be addressed. Significantly problematic are the lack of proper citation, insufficient length (the protocol part is one page too short) and the largely impersonal and distanced reflection.

For instance, when talking about challenging content you could deeper explore the reasons why it challenged you, maybe also in connection to your home study field. This is not about providing another sophisticated elaboration on the contents but may simply address the lecture structure, the presenters unique or lacking rhetoric style, whether the presentation matched your expectations of the topic or why and in what regards not? Did the topic resonate with you? Was it what you expected from the module or if not, what would you have liked to listen to/discuss? The reflection really is about your own individual perception and criticism. There is no right or wrong, just a scale from deep/insightful/reflective to superficial. Yours, at the moment, is rather the latter. That most sentences are subject to very generic word and syntax choices furthermore contribute to this impression.

Formal aspects also require improvement. The original version lacked proper section headings and was formatted left-aligned instead of justified. These structural elements are essential for meeting the pregiven academic standards.
A major concern is the absence of any adequate citation technique throughout the entire text. No in-text citations are provided, no bibliography is included, and no secondary literature (especially concerning the use of terminal vocabulary that demands explanation) is consulted or integrated into the discussion. Several statements require explicit references, for example:
- The description of the speaker's academic background ("The speaker's academic background lies in the history and philosophy of science…") should be supported by an official biography, institutional affiliation, or reference to relevant publications.
- The mention of historical devices such as the differential analyzer would benefit from clearer historical contextualization (e.g., dating, key figures, or literature references).
-Direct quotations such as "machines embodied mathematical assumptions" need proper citation to clearly mark them as paraphrased statements.

Please also note the comments that we left directly at exemplary text passages while you **revise your Protocol with Reflection**.
Without any text revision we unfortunately do not see your Protocol with Reflection meeting academic and formal standards.

Please also submit a signed Declaration of Academic Integrity (which you can find in moodle).

We are really looking forward for a more personal and more contextualized version of your submission!

Best wishes,
Team "Leonardo"

1. Protocol

The lecture was given in the context of the history of computing and science and addressed the reciprocal relationship between computer hardware development and scientific knowledge production throughout the 20th century. The speaker's academic background lies in the history and philosophy of science with a focus on computational practices, infrastructures, and the materiality of computing systems.

The lecture examined whether computers should be understood primarily as tools that accelerated existing scientific practices or whether they fundamentally transformed the nature of science itself. This question was explored historically by tracing developments from pre-digital calculation practices around 1900 to high-performance computing and GPUs, emphasizing how increasing hardware complexity reshaped scientific methods, models, and epistemic assumptions.

At the beginning, the lecture situated computers within a longer history of calculation. Around 1900, scientific computation relied heavily on **manual and semi-mechanical aids**, such as printed mathematical and logarithmic tables, slide rules, and other mathematical instruments. These tools structured scientific work by constraining both precision and complexity. The production and verification of tables themselves constituted a major scientific activity, highlighting that computation was already labor-intensive and socially organized before electronic computers existed.

By the 1930s, **mechanical and electromechanical machines** increasingly entered scientific practice. The lecture discussed mathematical machines such as the *differential analyzer*, which enabled the physical simulation of differential equations, and punched-card machines, which were used for large-scale data processing. These systems did not merely speed up calculations; they introduced new ways of representing problems and separating computational labor into distinct steps. One recurring point was that "machines embodied mathematical assumptions," meaning that hardware design already implied specific models of computation and problem structure.

The transition to electronic computers during and after World War II marked a qualitative shift. Early electronic computers enabled unprecedented calculation speeds and supported new types of numerical experimentation. The lecture emphasized that these machines were often developed in close connection with military, governmental, and large-scale research institutions. As a result, scientific questions increasingly adapted to what computers could calculate efficiently. According to the lecture notes, "the availability of computing time became a decisive factor in shaping research agendas."

The emergence of **mainframes and early supercomputers** further intensified this relationship. Large-scale simulations in physics, climate science, and engineering became

Formatted: Font: (Default) Arial, 11 pt, Bold

Formatted: Font: Bold

Formatted: Font: (Default) Arial, 11 pt

Formatted: List Paragraph

Formatted: Justified

Commented [CS1]: Prior to this sentence you should write a short paragraph that situates the lecture in the context of the "Leonardo" Project and the lecture series itself. That is, you should provide a brief outline of the earlier session topics and the general agenda/ theme of the lecture series. Also it is good practice to introduce the speaker and his work in the field to the reader and present how his talk (mention the title of the talk!) fits into that background.

Commented [CS2]: Here, you should provide references to his works. -> Author-Date in-text citations

Commented [CS3]: ?

Commented [CS4]: Where does the quote stem from?!

Formatted: Wave underline, English (United States)

Formatted: Wave underline, English (United States)

Commented [CS5]: This is no adequate way to cite a source. What do you mean by lecture notes? Whose quote is this? Where did it originate from? If you cite lecture slides, indicate so in parentheses with the slide number.

feasible, but only within institutional settings that could afford such machines. The lecture highlighted how access to computing resources, such as computing time at centralized facilities, influenced which research questions were pursued and which were postponed or abandoned. Computing infrastructure thus acted as a selective force within science.

A substantial part of the lecture focused on the rise of **minicomputers between the 1960s and 1980s**. These systems, such as commercially successful 12-bit minicomputers introduced around 1965 at comparatively low prices, decentralized computing power. For the first time, laboratories and research groups could operate their own computers without relying exclusively on national or institutional computing centers. This shift altered experimental practices, allowing tighter integration between data collection, analysis, and modeling. The lecture noted that minicomputers were widely used in scientific laboratories, including those associated with atomic energy and applied research.

The decentralization of computing was presented as a key factor in increasing **digital complexity**. As computers became more accessible, scientific workflows grew more computationally intensive and less transparent. Programs, numerical methods, and hardware configurations became integral parts of scientific results. One reproduced statement from the lecture captures this development: “Scientific results increasingly depended on software, hardware, and numerical approximations rather than closed-form solutions.”

Commented [CS6]: You still have to put in parentheses after the quote: (Hashagen, 2026: personal communication)

The lecture then moved toward late-20th-century developments, including **supercomputing and high-performance computing infrastructures**. These systems enabled large-scale simulations but also introduced new dependencies, such as the need for specialized knowledge in parallel programming and performance optimization. The example of computing time allocation at major computing centers illustrated how institutional policies could shape scientific outcomes indirectly by prioritizing certain projects or methods.

Finally, the lecture connected these historical developments to contemporary computing, briefly referencing **GPUs and deep learning** as an extension of earlier trends. While the title question—whether the computer drove science—was not answered conclusively, the lecture emphasized that hardware development and scientific practice co-evolved. Computers were neither neutral tools nor autonomous drivers of progress; instead, they functioned as **epistemic infrastructures** that structured what kinds of knowledge could be produced.

Commented [CS7]: These are incorrect dashes.

In summary, the lecture demonstrated that throughout the 20th century, advances in computer hardware repeatedly transformed scientific practice by enabling new forms of modeling, simulation, and data analysis while simultaneously introducing constraints related to access, cost, and technical complexity. The central result of the lecture can be summarized as follows: modern science cannot be understood independently of its computational material base, and the growth of digital complexity has fundamentally altered how scientific knowledge is generated, validated, and interpreted.

2. Reflection on the Lecture

Commented [CS8]: 80% of the text appear linguistically generic and impersonal. This is a safe space to learn writing coherent texts, make mistakes, correct yourself. We are not interested in your personal formulations, perspectives and thoughts

Formatted: Font: (Default) Arial, 11 pt, Bold

Formatted: List Paragraph, Justified, Numbered + Level: 1 + Numbering Style: 1, 2, 3, … + Start at: 1 + Alignment: Left + Aligned at: 0,63 cm + Indent at: 1,27

The lecture *"Did the Computer Drive Science? Hardware Development and Digital Complexity in the 20th Century"* formed a central and particularly illuminating contribution to the "Leonardo" Course on Digital Complexity. As an interdisciplinary course, the Leonardo series aims to question purely technical narratives of digital systems and instead situate them within broader cultural, historical, and epistemic frameworks. This lecture achieved that aim by critically examining the historical relationship between computing hardware and scientific knowledge production, thereby challenging common assumptions held within technical disciplines such as computer science. One of the most striking aspects of the lecture was its rejection of the idea that computers are merely neutral tools that accelerate pre-existing scientific practices. Instead, the lecture emphasized that hardware development and scientific inquiry have co-evolved throughout the 20th century. Computers were presented as *epistemic infrastructures* that actively shaped what kinds of scientific questions could be asked, how they could be answered, and which results were considered legitimate. This perspective stood out because it reframed technological progress not as a linear story of improvement, but as a series of historically contingent transformations that redefined scientific work itself.

Particularly important was the historical continuity drawn between pre-digital and digital computation. The discussion of logarithmic tables, mechanical calculators, and early mathematical machines demonstrated that computation has always been embedded in material practices and social organization. Scientific labor around 1900 involved not only theoretical reasoning but also the production, verification, and maintenance of computational artifacts such as tables. The lecture's statement that "machines embody mathematical assumptions" highlighted how even early computational tools constrained scientific reasoning by privileging certain representations and methods. Analytically, this challenges a simplified distinction between "theory" and "tool" and instead suggests that material devices participate directly in knowledge production.

Commented [EK9]: Above you cited: "machines embodied mathematical assumptions". Only one of these versions can be the original one. You may adapt a quote to fit the grammar of your sentence but then any changes need to be put in these brackets […]. Make sure to quote correctly and to highlight if it is a direct quotation from the professor or someone else!

From my perspective as a computer science student, this historical framing was especially valuable because it provided a broader context for phenomena that are often taken for granted today. For example, the lecture showed that the rise of electronic computers during and after World War II did not simply increase speed and efficiency, but fundamentally altered scientific epistemology. Numerical simulation and computational experimentation emerged as legitimate modes of inquiry alongside theory and experiment. The availability of computing time, access to centralized mainframes, and later the decentralization brought about by minicomputers shaped research agendas in subtle but powerful ways. The observation that

Formatted: Justified

"the availability of computing resources became a decisive factor in shaping research agendas" resonated strongly, as it remains relevant in contemporary contexts such as cloud computing and large-scale AI models.

Commented [CS11]: Whose quote is this? Where did it appear?

Regarding my prior knowledge, the lecture was largely accessible due to my technical background, yet it required a deliberate shift in perspective. Concepts related to hardware evolution, numerical methods, and high-performance computing were familiar, but they were embedded in an analytical framework that emphasized historical causality and epistemic consequences rather than optimization or performance. This made the lecture intellectually demanding in a constructive way. Rather than introducing unfamiliar technical content, it challenged familiar concepts by placing them in an unfamiliar interpretive context. As a result, following the lecture required careful listening and reflection rather than technical problem-solving.

My ability to contribute to the discussion was influenced by this interdisciplinary setting. I felt confident contributing when the discussion touched on contemporary computing topics, such as GPUs, machine learning, and the increasing opacity of computational systems. In these moments, I could connect historical arguments to present-day technical practices. However, I was more cautious when discussion focused on historical interpretation or theoretical concepts from science and technology studies. This hesitation reflected a difference in disciplinary training rather than a lack of interest. It also illustrated a broader issue in interdisciplinary courses: meaningful participation often requires not only shared topics but also familiarity with different argumentative styles and evaluative criteria.

Commented [EK12]: What specifically appeared demanding to you regarding the historical interpretations and theoretical concepts?

In terms of insights gained for my own studies and future career, the lecture encouraged a more reflective understanding of technical work. It became clear that developing computational systems is not a purely neutral or technical activity, but one that shapes how knowledge is produced, validated, and trusted. For instance, the increasing reliance on simulations and data-driven models implies a shift away from analytical transparency toward computational plausibility. This has implications for reproducibility, explainability, and scientific accountability—issues that are becoming increasingly important in fields such as artificial intelligence and data science. The lecture thus reinforced the idea that technical expertise should be complemented by historical and ethical awareness.

Commented [CS13]: See above..

Looking at my own discipline, I recognized that computer science often treats hardware and infrastructure as given constraints rather than as historically and socially shaped entities. While I had previously encountered discussions about computational limits, numerical errors, or algorithmic complexity, these topics were typically framed as engineering challenges rather than epistemic concerns. The lecture made me aware of this limitation: computer science excels at abstraction, but this strength can also obscure the broader consequences of computational practices. Recognizing this limitation does not weaken the discipline, but rather highlights the need for interdisciplinary engagement.

At the same time, the lecture offered valuable insights into the foreign discipline of history and philosophy of science. I gained a deeper appreciation for how historical analysis can uncover long-term structural patterns, such as the relationship between institutional power and access to computing resources. However, I also became aware of potential limitations. Historical narratives may sometimes underrepresent the internal technical constraints that shape design decisions, such as performance trade-offs or algorithmic feasibility. This observation further underscored the importance of dialogue between disciplines rather than prioritizing one perspective over another.

Several critical points emerged that I would like to explore further. One concerns the growing complexity and opacity of computational systems and the resulting reliance on trust in both hardware and software. Another concerns the political and economic dimensions of computing infrastructure, particularly the concentration of computational power in large institutions or corporations. These issues seem highly relevant not only for the Leonardo Course but also for broader debates about digital society and governance.

Finally, the lecture fit exceptionally well within the overall concept of the Leonardo Course on Digital Complexity. It complemented previous sessions by providing a concrete historical case study that illustrated how digital complexity emerged over time and why it cannot be understood solely in technical terms. By linking historical developments to contemporary issues, the lecture reinforced the course's central message: digital complexity is not just a property of systems, but a condition of modern knowledge production.

Commented [CS14]: If you say "exceptionally", it would be necesssary and an an intersting insight as to how some other lectures fit less well in the context - which is a correct observation - but you should elaborate on why, in your own words.

In conclusion, the lecture deepened my understanding of the intertwined histories of computing and science while also prompting critical reflection on my own discipline. It demonstrated that to engage responsibly with digital complexity, technical expertise must be accompanied by historical, epistemic, and interdisciplinary awareness.

Works Cited section missing

Formatted: Justified